Islamic University of Technology

Lab 4: Expressions and Math

CSE 4108 : Structured Programming Lab

Section 1B

MIA, NFS, IAC, ASH

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1 Lab Tasks

1.1 Reversed Plus One

Write a C program that uses takes input a four digit integer from the user and then prints the digits reversed with 1 added to each digit.

[Sample Program]
Input

```
Enter 4 digit integer: 1342
```

Output

```
Reversed 4 digit integer: 3542
```

[Hint] *If n is an integer, then n%10 is the last digit in n and n/10 is n with the last digit removed.*

1.2 Birth of a Cult

Card User Land Team (CULT) has a secret code number that it assigns for each new member. Cult members are avid followers of astrology, so every member's code can be calculated from the exact time of their birth. The time of birth is in the format `Hh Mm Dd Xx PQRS` (hour, minute, day, month, year).

The formula to calculate code from time of birth is to multiply D with d and M with m, add them with P+Q+R+S. Then H is subtracted from the answer and h+X is added to it. Finally the remainder of the answer with respect to x is the code.

[Sample Program]
Input

```
Enter Birth Time & Date: 15 30 09 12 2001
```

Output

```
Code: 0
```

[Hint] *If a number 123456 is scanned with* `scanf("%3d%2d%d", &a, &b, &c)`*, then* `a` *will be 123,* `b` *will be 45 and* `c` *will be 6. In other words, in* `scanf`*,* `%nd` *scans the first* `n` *digits and leaves the next digits for the next specification(s).*

1.3 Swapper

Write a C program that would take two integers as input from the user and store then in variable `a` and `b`. Then the program will swap the values of the variables `a` and `b`, so that `b` has the value of `a` and vice versa. Then print out the swapped values. Swap them in the variables. Do not just try to print `b` first and then `a`.

[Sample Program] *Input*

```
Enter two integers: 10 15
```

Output

```
Swapped integers: 15 10
```

[Hint] *If you have one glass of orange juice and one glass of milk, you cannot swap the liquids of both the glasses without the help of a third glass.*

1.4 Long distance Education

Larry Potter sends an owl with his answer scripts to his lecturer after every exam. Through magic, Larry knows the initial velocity `u`, final velocity `v` and acceleration `a` of the owl. Help Larry calculate the distance `s` his owl needs to travel. Also calculate the time it would take `t`.

[Sample Program]
Input

```
Enter Initial Velocity (ms): 10.0
Enter Final Velocity (ms): 44.22
Enter Acceleration (a): 1.2
```

Output

```
s: 773.08683 meters
t: 28.51667 seconds
```

[Hint] *Google "Kinematics Formulas" to find the required equations. Use* `<math.h>` *library for the required functions,* `sin`, `cos` *and* `pow`.

1.5 Variance

Write a C program that would take two float numbers as input and print out their Variance.

[Sample Program]
Input

```
Enter number A: 5.0
Enter number B: 15.0
```

Output

```
Variance: 25.00000
```

[Hint] *Google "Variance between two numbers" to find the required equation. Use* `<math.h>` *library for the required functions,* `sqrt` *and* `pow`.

2 About Math Library

C Math library contains useful functions for common mathematical calculations. The functions can be used by including the header file `<math.h>`. A reference to all the functions in math library can be found at "https://en.cppreference.com/w/c/numeric/math".